JASPER LEE

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EDUCATION

The University of Illinois Urbana-Champaign, Urbana, IL — May 2030
PhD in Computer Science — GPA: 4.00/4.00

The University of Texas at Austin, Austin, TX — May 2025
BS in Computer Science, Turing Scholars Honors Program — GPA: 3.98/4.00
BS in Mathematics

RESEARCH EXPERIENCE

Formally Certified Automation and Learning Lab, Urbana, IL (Advisor: **Prof. Gagandeep Singh**) — January 2026 - Present

- Doing research at the intersection of formal methods and ML
- Developing post-training methods for LLM assistants in multi-turn conversations with tool calls, ensuring they follow corresponding formal specifications
- Exploring various ways to incorporate formal logics (such as linear temporal logic) into ML training and evaluation

Illinois Theorem Provers Lab, Urbana, IL (Advisor: **Prof. Talia Ringer**) — August 2025 - Present

- Doing research on improving ML for formal theorem proving and LLMs for code reasoning
- Exploring novel fine-tuning paradigms for improving LLMs' code reasoning (e.g. output prediction) capabilities
- Testing various techniques for training ML models to emulate fundamental structurally recursive algorithms
- Exploring a growing interest in formally verified LLM sandboxing using information flow

Trustworthy Intelligent Systems Lab, Austin, TX (Advisor: **Prof. Swarat Chaudhuri**) — August 2023 - May 2025

- Did research in automated theorem proving, LLM agents, and reinforcement learning
- Created, tuned, and evaluated an automated theorem prover that uses a series of LLM agents to generate Lean 4 programs and corresponding formal correctness proofs for the CLEVER benchmark (NeurIPS 2025)
- Helped create PutnamBench, an automated theorem proving benchmark of 640 problems from the Putnam Mathematical Competition, written in the proof assistants Lean 4, Isabelle, and Coq; the PutnamBench paper was accepted at NeurIPS 2024 and won the Best Paper Award at the AI for Math Workshop @ ICML 2024
- Worked on solving reinforcement learning problems with high reward by using pre-trained LLMs to choose actions

PROFESSIONAL EXPERIENCE

Axiom Math Research Internship, Palo Alto, CA — May 2026 - August 2026

- Developed novel automated theorem proving methods for improving Lean theorem proving performance

Jane Street Capital Software Engineering Internship, New York, NY — May 2024 - August 2024

- Implemented two software engineering projects in OCaml and Python, working with databases and data manipulation
- Presented both projects to my teams at the end

PUBLICATIONS

Amitayush Thakur, **Jasper Lee**, George Tsoukalas, Meghana Sistla, Matthew Zhao, Stefan Zetzsche, Greg Durrett, Yisong Yue, and Swarat Chaudhuri. **CLEVER: A Curated Benchmark for Formally Verified Code Generation**. In ***NeurIPS 2025*** *Datasets and Benchmarks Track*, 2025. URL https://openreview.net/forum?id=IbOacMF5qd.

George Tsoukalas, **Jasper Lee**, John Jennings, Jimmy Xin, Michelle Ding, Michael Jennings, Amitayush Thakur, and Swarat Chaudhuri. **PutnamBench: Evaluating Neural Theorem-Provers on the Putnam Mathematical Competition**. In ***NeurIPS 2024*** *Datasets and Benchmarks Track*, 2024. URL https://openreview.net/forum?id=ChKCF75Ocd.

HONORS AND AWARDS

NSF Graduate Research Fellowship — Fall 2025 - Spring 2030

- National fellowship program for graduate STEM researchers, awarded $53,000/year for 3 years

Best Paper Award, AI for Math Workshop @ ICML 2024 — June 2024

- Best Paper Award at the AI for Math Workshop @ ICML 2024 for the paper "PutnamBench: A Multilingual Competition-Mathematics Benchmark for Formal Theorem-Proving"

SKILLS

Programming Languages: Python, C, C++, Java, Rust, Go, OCaml, Rocq, Lean 4, MATLAB, R, C#, x86/ARM assembly, Verilog
Technical/Computer Skills: PyTorch, TensorFlow, Jupyter Notebooks, Git, LaTeX, Linux, SQL, Unity
Additional Skills: Technical writing, Presentation